

**Geotechnical Investigation
Proposed Projects Office**
Stanwell Power Station, Stanwell

Prepared for
Stanwell Corporation Limited
Project No. RX25-1156A

10 December 2025

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Important Information about your Geotechnical Engineering Report (2 pages)

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Drawing No. 1	Locality Plan and Bore Locations
Appendix A	Bore Report Sheets with Explanatory Notes
Appendix B	Laboratory Test Result Report Sheets

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SECTION 1 INTRODUCTION

1.1 Project

It is understood that geotechnical investigation is required for the design and construction of a proposed Projects Office at the Stanwell Power Station, Stanwell. The proposed new development is likely to include a new single-storey building to replace the existing projects office buildings. The location and extent of the proposed Project office are indicated approximately on Drawing No. 1.02, attached.

1.2 Proposed Scope of Work

It was requested by Stanwell Corporation Limited (Stanwell) that geotechnical investigation (in conjunction with the contamination assessment) of the site comprise the drilling and sampling of two bores to 6m depth at suitable locations within the building footprint. Using the results of the proposed fieldwork and laboratory testing outcomes, Stanwell requested a report to be produced that would provide geotechnical design information on each of the following topics, as appropriate:

- subsurface conditions;
- groundwater observations;
- earthworks and site preparation recommendations;
- advice on suitability of using existing material as fill;
- erodability and erosion control parameters;
- stability of temporary and/or permanent slopes including temporary and permanent cut/fill batter angles;
- advice on recommended foundation types;
- high level footings working bearing pressures and recommended founding strata;
- deep foundation working capacities;
- reactive ground movement and site classification to AS2780: *Residential Slabs and Footings*;
- elastic foundation settlement estimates;
- slab on ground and pavement geotechnical design parameters;
- site classification in accordance with AS1170.4 for earthquake action;
- anticipated construction aspects; and
- any other relevant geotechnical parameters.

The preliminary contamination investigation is reported separately.

1.3 Commission

Based on the proposed scope of work provided by Stanwell, undertaken in conjunction with the contamination investigation, a fee to undertake the investigation was presented in a proposal dated 6 May 2025. Butler Partners Pty Ltd (Butler Partners) was subsequently commissioned by Stanwell Corporation Limited to conduct the geotechnical investigation as proposed. A draft report was provided for comment on 31 July 2025.

SECTION 2 THE SITE

2.1 Background

Butler Partners has undertaken previous geotechnical investigation in Future Energy Innovation and Training Hub (FEITH), approximately 200m north-east of the Projects Office site and the results are given in the following report:

Geotechnical Investigation
Proposed FEITH Iron Battery Project
Project No. RG24-1468B
29 May 2024

The investigation comprised a surface layer of gravelly/sandy/silty clay fill to between 0.2m and 0.5m and up to 2.0m depth. Underlying the fill was hard silty clay to between 1.5m and 3.5m, underlain in turn by weathered mudstone/sandstone rock of extremely low to low strength

2.2 Site Description

The proposed Project office is located approximately 350m north-east of the existing generators within Stanwell Power Station and adjacent to the existing training facilities building and security/administration building further to the south. The main roadway to enter the site is located immediately east of the development area, with the main access road comprising a bus stop and visitor parking area.

At the time of the investigation, the proposed development site was occupied by a raised building in the southern portion, and a car park and outdoor lunch area in the northern portion. Garden beds, comprising small to medium mature trees/shrubs, enclosed the north and eastern sides of the site, with a large shade cloth in the northeastern corner. Vegetation across the vacant portions of the development area comprised sparse grasses, however, areas of bare earth were also observed. A covered shipping container on a concrete pad occupied a small area on the eastern side adjacent to the existing building.

Ground surface levels across the development site were generally uniform at approximately RL62m. An aerial image of the proposed development site highlighted in blue, is presented in Photograph 1, and general views of the bore4 and bore1 at the time of the investigation are presented in Photographs 2 and 3.



Photograph 1: Aerial image showing the approximate Project Office site extent (Nearmap Image, 4 July 2025)



Photograph 2: General view of the proposed Projects Office looking north-west towards Bore 4.



Photograph 3: General view of the site looking south-east towards Bore 1.

2.3 Geology

Reference to Queensland Globe Geology Mapping and the Australian Stratigraphic Units Database indicates that the site is mapped within areas of the Early Cretaceous aged Stanwell Formation, consisting of mudstone, sandstone, siltstone and minor coal.

SECTION 3 FIELDWORK

3.1 Drilling and Sampling Methods

The investigation comprised the drilling and sampling of two bores (Bores 1 and 4) to approximately 6m depth. The bores were drilled with a truck-mounted Jacro 350 drilling rig, using solid flight auger drilling methods. Strata identification was based on inspection of cuttings recovered on the augers and 'disturbed' Standard Penetration Test (SPT) samples, recovered at selected depths.

On completion of drilling, the bores were backfilled with drill spoil and surface plugged.

Bores 2 and 3 were nominated as environmental bores only, completed as part of the preliminary contamination investigation, and have been included in this report for reference.

3.2 Bore Locations and Supervision

The bores were set out in the field by direct measurement from existing site features and are indicated approximately on the attached Drawing No. 1.02. An experienced geotechnical engineer logged the stratigraphy encountered in the bores, directed the in-situ sampling and testing program and supervised the fieldwork.

SECTION 4 INVESTIGATION RESULTS

4.1 Subsurface Conditions

The subsurface conditions encountered are given on Bore Report sheets included in Appendix A, using classification and descriptive terms defined in the accompanying notes (which are generally in accordance with Australian Standard AS AS1726-1993). It should be noted that rock types indicated on the Bore Report sheets are based on visual assessment only; no petrographic analysis has been undertaken for confirmation.

For a description of the stratigraphy encountered at each bore location, the Bore Report sheets should be consulted. However, in broad summary, the subsurface conditions encountered at the bore locations generally comprised a surface layer of silty gravel/silty sand fill to 0.1m and 0.2m depth, underlain by very stiff to hard sandy/silty clay to 4.5m, underlain in turn by extremely low strength siltstone/sandstone rock.

4.2 Groundwater

Free groundwater was not encountered during auger drilling of any of the bores, however, groundwater levels would be expected to be affected by seasonal and prevailing weather conditions. If construction is to be undertaken at a significant time following this investigation and/or following significant wet weather, it would be prudent to confirm groundwater levels prior to construction.

4.3 Laboratory Testing

Selected samples of soil and extremely weathered rock recovered from the bores were submitted to Butler Partners' NATA accredited Rockhampton geotechnical testing laboratory for assessment of erosion and sediment control parameters, particle size distribution and plasticity using Australian Standards AS1289 test methods. Laboratory test report sheets are included in Appendix B and the test results are summarised in the following sections.

It should be noted that sample descriptions provided in the laboratory results summary tables (and the laboratory test result sheets) are based on the inspection of each individual laboratory test sample only. No allowance has been made in sample descriptions for sampling, sub-sampling or test methodology in determination of the mass material properties. Estimates of mass material properties are provided on each individual Bore Report sheet and as such, the laboratory test results should be read in conjunction with the relevant report sheets.

4.3.1 Erosion and Sediment Control Parameters

Two selected samples of soil were tested to determine the Emerson Class Number, pH and electrical conductivity. A summary of the reported test results is presented in Table 1 and the Emerson Class Number results indicate that the samples tested had a low potential for dispersion, using distilled water.

Table 1: Summary of Reported Emerson Class, pH and Conductivity Test Results

Bore	Depth (m)	Sample Description	Sample Moisture Content (%)	Emerson Class No.	pH	Electrical Conductivity (µS/cm)
1	0.1 – 0.5	Silty Clay	8.1	4	8.6	170.7
4	0.2 – 0.5	Silty Clay	11.3	4	9.2	455.3

4.3.2 Particle Size Distribution

Three selected samples of soil were tested for measurement of particle size distribution analysis using wash sieve grading techniques, and the reported results are summarised in Table 2.

Table 2: Summary of Reported Particle Size Distribution Test Results

Bore	Depth (m)	Sample Description	Sample Moisture Content (%)	Gravel Fraction ⁽¹⁾ (%)	Sand Fraction ⁽²⁾ (%)	Silt and Clay Fraction ⁽³⁾ (%)
1	0.95 – 1.5	Silty Clay	12.9	9	13	78
	3.27 – 4.5	Silty Clay	14.4	9	13	78
4	0.95 – 1.5	Silty Clay	11.3	6	24	70

⁽¹⁾ Particle size <60mm, >2mm; ⁽²⁾ Particle size (approximately) <2mm, >0.075mm; ⁽³⁾ Particle size (approximately) <0.075mm

4.3.3 Plasticity

Three selected samples of soil and extremely weathered rock were tested for measurement of plasticity using Atterberg limit and linear shrinkage test methods. The reported test results are summarised in Table 3 together with the sample classification and indicate that the samples tested were predominantly of medium to high plasticity.

Table 3: Summary of Reported Plasticity Test Results

Bore	Depth (m)	Sample Description	Sample Moisture Content (%)	Liquid Limit (%)	Plastic Limit (%)	Plasticity Index (%)	Linear Shrinkage (%)	Classification ⁽¹⁾
1	0.95 – 1.5	Silty Clay	12.9	69	18	51	9.5	CH
	3.27 – 4.5	Silty Clay	14.4	60	16	44	15.0	CH
4	0.95 – 1.5	Silty Clay	11.3	43	16	27	11.5	CI

⁽¹⁾ Australian Standard AS1726-1993, *Geotechnical site investigations*

SECTION 5 GEOTECHNICAL DESIGN DISCUSSION

5.1 Ground Model

The results of the investigation indicate that the site was underlain at the bore locations by a surface layer of silty gravel/silty sand fill to between 0.1m and 0.2m depth, underlain by sandy/silty clay to 4.5m, underlain in turn by extremely low strength siltstone/sandstone (rock). Groundwater was not encountered in any bore; however, groundwater levels can change over time. In these ground conditions geotechnical design will need to consider (at least) the following:

- subsurface conditions;
- excavatability;
- site trafficability;
- suitability of cut for fill;
- erosion and sediment control;
- batter slopes;
- reactive soil movements;
- suitable foundation types;
- working bearing pressures;
- potential for strength inversions to influence foundation design;
- foundation settlement estimates;
- pavement and slab subgrade properties;
- site earthquake classification; and
- possible construction difficulties.

Discussion of geotechnical design parameters, as well as design and construction recommendations and suggestions are detailed in the following sections.

5.2 Earthworks

5.2.1 Existing Fill

Existing fill was encountered in the majority of the bores (generally less than 0.2m depth), and if required to support new structural fill, on-ground slabs, pavements or other settlement sensitive elements of the development, supporting documentation should be obtained and checked to confirm that the fill has been placed in a controlled manner to a specification that is appropriate for the proposed development. If documentation does not exist (or the specification used for filling is not appropriate), then it is suggested that the existing fill be assumed to be 'uncontrolled', which would be consistent with the appearance of the fill in the bores.

If fill cannot be shown to be controlled (to an appropriate specification), then consideration should be given to the potential for adverse variation to exist in both the composition and degree of compaction of the fill. The presence of voids within uncontrolled fill as well as potential soft/loose zones or inclusions of deleterious material and cobbles/boulders, etc. may lead to potentially significant total and differential settlements under the action of structure loads, occurring possibly over relatively short distances.

To minimise the risk of potentially adverse settlement within uncontrolled fill due to poorly compacted zones or inclusions of deleterious material occurring, it is recommended that the following options be considered;

- settlement sensitive structures/features should be fully supported on foundations extended to a suitable material below the fill, or
- where relatively 'lightly loaded' structures are proposed to be founded (shallow) in the existing uncontrolled fill, they should be either sufficiently designed and detailed to tolerate potentially significant total and differential settlement in the long term, or
- the existing uncontrolled fill be removed and recompacted to an appropriate standard (under Level 1 conditions).

5.2.2 Site Preparation

Site preparation in areas of future settlement sensitivity should involve removal of any uncontrolled fill, unsuitable materials, vegetation, topsoils, etc., as far in advance of construction as possible.

Following site stripping (and excavation, as required), the exposed subgrade in areas to be developed should be uniformly compacted to the relevant (Standard) dry density ratios indicated in Table 4. 'Soft spot' areas may be encountered in exposed clay subgrade (particularly if earthworks are undertaken during and following 'wet' weather) and potentially beneath any existing site fill and ponded water.

Care will be required to ensure that the effect of site earthworks does not impact adversely upon adjacent services and structures etc. (e.g. as potential settlements induced by vibratory compaction, etc.).

5.2.3 Excavatability

Excavation of fill, soils and extremely low strength rock should be readily achieved in confined and bulk excavation using a large hydraulic excavator. It is also expected that drilling of bored piles in the weathered siltstone/sandstone (rock) would be achievable with a large hydraulic excavator or medium to large sized piling rig.

All confined excavations should be fully supported or battered to ensure personnel safety where entry is required.

It should be noted that 'harder' rock may exist close below bore termination depths and at shallower depth elsewhere on the site.

5.2.4 Suitability of Excavated Materials for Reuse as Fill

Except for soils that are organic, over-wet or contain over-size (i.e. >75mm) particles, materials recovered in excavations should be suitable for re-use as fill, provided expansive behaviour can be tolerated or designed for where reactive materials (e.g. sandy/silty natural clays) are used.

The clays encountered in the bores are considered to generally have potential for high shrink-swell movements associated with change in moisture content, especially (but not only), if they are recompacted. It should be noted that heavily compacting reactive fill (particularly at a moisture content below Standard OMC) should be avoided as potentially significant expansion could occur on 'wetting up'.

5.2.5 Fill Compaction

All fill supporting settlement sensitive structures/features should be 'controlled' and placed in layers having a loose thickness of not greater than 250mm and uniformly compacted to the minimum dry density ratio values nominated in Table 4.

Table 4: Subgrade and Fill Compaction Levels

Description	Minimum Dry Density Ratio
General Fill (non-structural support)	95% (Standard compaction)
Floor slab and foundation support	100% (Standard Compaction)
Pavement Subgrade – >500mm below subgrade level	98% (Standard compaction)
– top 500mm of subgrade	100% (Standard compaction)

Reactive clay material should be avoided for use as fill, where possible. However, if its use cannot be avoided, any reactive material used as fill should be placed and maintained at a moisture content of not less than Standard OMC in order to reduce potential shrink-swell movements. Due allowance must be made in design and detailing for reactive fill movements, if reactive fill is used.

To assist with achievement of adequate control of fill placement, geotechnical testing as set out in Section 8 of AS3798 – 2007 *Guidelines on earthworks for commercial and residential developments* is suggested. Where fill is to be placed beneath pavement areas, 'Level 2' geotechnical supervision and testing could be adopted, and where fill is to be placed beneath buildings and/or movement-sensitive areas, 'Level 1' supervision is recommended.

5.2.6 Trafficability

Site trafficability would be difficult to impossible for rubber tyred plant in the 'exposed' subgrade clays if 'wet' and a granular trafficability layer would be required to provide all-weather site access. The thickness of the trafficability layer must be determined by analysis for the specific type of plant that will traffic the site. However, for light weight plant (only) a minimum thickness of not less than 0.2m of coarse granular material would be expected.

5.2.7 Site Drainage

During construction, the site should be graded such that water is readily shed and does not collect and pond over the site, otherwise softening of soil, fill and weathered rock subgrade will occur, especially under construction plant traffic and heavy vehicles.

5.2.8 Erosion and Sediment Control

The results of the laboratory testing indicate that the samples tested (refer Table 1) had a low potential for dispersion (Emerson Class No.4). The results of testing given in Table 1 can be used, in conjunction with relevant soil classifications from the Bore Report sheets, to input to an erosion hazard assessment for the site (e.g. IECA, 2008¹).

¹ IECA (2008), *Best Practice Erosion and Sediment Control*, International Erosion Control Association (Australiasia), Picton NSW – Book 4

5.3 Batter Slopes

Temporary and permanent batter slopes up to 3m high (unsurcharged by structures etc.) could be formed at the angles given in Table 5, subject to confirmation by location specific stability analysis and inspection during construction. At the batter slopes given, some movements at and behind the slope crest would be anticipated.

Table 5: Maximum Unsurcharged Temporary and Permanent Batter Slopes up to 3m high.

Material	Strength	Temporary Batter ^{(1) (2)}	Permanent Batter ^{(1) (2)}
Level 1 Controlled Fill	–	1V:1H	1V:2.5H
Sandy/Silty Clay	stiff to hard	1V:1H	1V:2H
Mudstone/Sandstone (rock)	extremely low	1V:1H ⁽³⁾	1V:2H ⁽³⁾

⁽¹⁾ Not underlain by 'softer' materials and subject to confirmation by engineering analysis and inspection during construction

⁽²⁾ Flatter if 'wet'

⁽³⁾ Depends on jointing

At the batter angles nominated in Table 5 there may be some localised slumping of batter slopes and it will be necessary to ensure that the faces are protected from any surface water or groundwater seepage effects (e.g. by grading sites away from slopes and/or the use of surface and subsurface drainage lines).

All batter slopes excavated (and any fill batters) should be protected from weathering, erosion and groundwater seepage and should be inspected by an experienced engineer during earthworks to confirm stability and revise batter slopes, if required.

5.4 Reactive Ground Movements

5.4.1 Magnitude

The magnitude of potential reactive soil movements can be estimated using the following equation (from Australian Standard AS2870-2011 *Residential slabs and footings*) and parameters for the site selected based on recommendations in AS2870:

$$y_s = \frac{1}{100} \sum_{n=1}^N \left(\alpha \cdot I_{ss} \cdot \overline{\Delta u} \cdot h \right)_n$$

where y_s is the characteristic surface movement, in millimetres;
 α is the lateral restraint factor;
 I_{ss} is the shrink-swell index taken as 3.0 per pF to 3.5% per pF for the site clays based on past experience, and the shrink-swell index test results taken from the investigation completed for the FEITH project (approximately 200m north-west);
 $\overline{\Delta u}$ is the soil suction change averaged over the thickness of the layer under consideration (taken as 1.5pF at the ground surface in Stanwell and reducing linearly to zero at the design depth of suction change);
 h is the thickness of layer under consideration, in millimetres; and
 N is the number of soil layers within the design depth of suction change (H_s), which has been taken as 2.3m in Stanwell.

Based on an assessment of the materials encountered in the bores, a shrink-swell index taken as 3.0 per pF to 3.5% per pF, and using the methods described in Australian Standard AS 2870, it is estimated that free ground surface movements as a result of soil shrinking and swelling from normal soil moisture variations for the site in its current condition could vary between 49mm and 57mm.

5.4.2 Site Classification

Notwithstanding the calculated characteristic surface movement values, due the presence of uncontrolled fill encountered at the site would be classified as Class 'P'; if the existing fill was removed (which is considered to be readily achievable), the site could be classified as 'Class H1' (Highly Reactive).

5.4.3 Design Considerations

The clays encountered at the site are expected to generally have high potential for shrink-swell movements associated with change in moisture content. If recompacted, the potential reactivity of these materials will increase, which will need to be allowed for in the estimation of future shrink-swell movement.

Use of reactive materials for fill should be avoided, however, if their use cannot be avoided then the calculated characteristic surface movement value would increase significantly. It should be carefully noted that the calculated surface movement values given above do not include any allowance for 'abnormal' influences such as vegetation effects. It is strongly recommended that the estimated characteristic surface movement values for the site be recalculated once site earthworks design is completed and fill sources are known. It is considered that the following issues must be carefully considered in design:

- Where reactive fill is placed over a natural soil subgrade, higher characteristic movements than those nominated above could potentially occur (as the ratio of lateral restrained to unrestrained movement will increase), particularly if the fill reactivity is greater than that of the existing site soils. If filling of the site is proposed, a revised site classification should be considered, which takes into account the actual reactivity, compaction and depth of fill used.
- Vegetation (particularly large trees) has the potential to significantly increase soil suction change magnitude and depth, which leads to a significant increase in potential reactive soil movements adjacent to any (proposed) tree locations. If trees are to be planted 'close' to proposed structures in the future, consideration should be given to constructing root barriers around the trees, and footing design must allow for potentially (significantly) higher reactive soil movements than are nominated above.
- Abnormal subgrade moisture variations could potentially result in adverse, non-uniform reactive movements that are significantly greater than those nominated above for 'normal' seasonal moisture changes. The risk of 'abnormal' movement occurring could be reduced by ensuring over-watering of gardens, ponding water, broken/leaking pipes, 'close' planting of trees/shrubs, etc. does not occur.

'Good practice' should be adopted in project design and detailing if control of reactive ground movement is desired. In particular, the following are recommended:

- trees/shrubs should not be planted closer than their mature height to movement sensitive features (unless significantly greater reactive movements than those estimated above are designed for);
- subgrade moisture content should not be allowed to change during or following construction;
- site grades should be designed to readily shed water and prevent ponding around footings and other movement sensitive areas;
- services should be designed to be flexible and to prevent any leakage and to rapidly promote removal of fluid if leakage does occur; and
- proposed structures should be made as flexible as possible, with regular full height movement control joints, flexible in-fill above windows and doors etc.

5.5 Foundations

Based on the results of the bores, either strip and pad footings could be used to support proposed structures in the very stiff to hard clays, provided that the structures can be designed for, or tolerate, reactive ground movement due to soil moisture content/suction change. For 'heavier' structures sensitive to ground movement, deep foundation systems (i.e. bored piles) supported on weathered sandstone/siltstone (rock) could be considered.

It is considered that local variations in soil/rock strength could be expected to occur over the site and it is suggested that the foundation design and construction pricing/methodology adopted for the project be such that footing sizes/founding depths can be readily adjusted if required during construction, without significant cost/time penalties being incurred.

In order to minimise potential differential foundation settlements, all footings to individual structures should be fully supported in the same quality material underlying the structure, unless each structure can be designed and detailed to be relatively flexible (e.g. footings to individual structures should not be supported in both clays and rock).

5.5.1 Strip and Pad Footings

Maximum allowable working bearing pressures are given in Table 6 for foundation sizing. Ultimate (failure) bearing stress design values can be estimated by multiplying the working stress bearing pressure values by 2.5.

Table 6: Maximum Working Stress Design Parameters for Strip and Pad Footings

Material	Strength ⁽²⁾	Maximum Working Bearing Capacity ⁽¹⁾ (kPa)
Level 1 Controlled Fill	-	100
Sandy/Silty Clay	very stiff	150
	hard	250
Siltstone/Sandstone (rock)	extremely low	450

⁽¹⁾ Not underlain by 'softer' material;

⁽²⁾ Refer to attached soil and rock classification sheets

All foundation excavations should be inspected by an experienced geotechnical engineer prior to casting, to confirm adequate bearing capacity (and final foundation dimensions/depths) and all footing excavations should be clean, dry and free of softened materials immediately prior to casting. If footing excavations are to remain open for longer than ten hours prior to casting (which is not recommended) and/or wet weather is likely, it is suggested that a concrete blinding layer be cast in the base to prevent moisture ingress (and base softening).

If strip and pad foundations are adopted in or through reactive materials, it will be necessary to carefully assess if moisture/suction change can occur in the reactive materials after construction and whether allowance in design is required to minimise uplift forces/movement on footings and/or suspension of ground beams etc. on 'void former'. Conventional design options to minimise reactive ground movement effects on strip and pad footings/ground beams comprise one or a combination of the following:

- increasing founding depths to reduce uplift movement;
- surrounding structures with an appropriate depth, impermeable vertical cut-off wall to restrict moisture content/suction change in reactive founding materials;

- excavation and replacement (to an appropriate depth), of reactive founding material, with non-reactive replacement fill to reduce uplift movement;
- fully surround the exterior of structures with an impermeable 'apron' of not less than 3m width (but to be determined on a location by location basis), to minimise moisture content/suction change in the reactive founding materials; and/or
- place and maintain reactive fill 'wet' of Standard Optimum Moisture Content (OMC).

5.5.2 Raft Foundation

It is considered that the very stiff to hard clays encountered in the bores may be suitable to support a raft foundation, subject to the required working bearing pressure, and the distribution of structural load over the site. If any 'soft spots' or firm layers are encountered at raft subgrade level, they will probably require excavation and replacement with Level 1 controlled select fill; geotechnical assessment would be required to confirm the uniformity of the soils at raft founding level and the extent/depth of any 'soft' areas.

For the preliminary assessment of raft performance, the parameters presented in Table 7 could be used, which have been based on the limited results of this investigation, past experience and published correlations. Detailed analysis will be required to confirm that a raft option is feasible for any section of the development.

Table 7: *Estimated Settlement Modulus and Poisson's Ratio Values*

Material	Strength	Estimated Settlement Modulus (MPa)	Poisson's Ratio
Level 1 Controlled Fill	-	20 – 50	0.3
Sandy/Silty Clay	very stiff	20 – 40	
	hard	40 – 70	
Siltstone/Sandstone (rock)	extremely low	100 – 150	0.25

⁽¹⁾ Not underlain by 'softer' material

If the existing ground strength/stiffness is not sufficient to support a raft foundation at the required working bearing pressure, consideration could be given to installing piles beneath the raft to form a piled-raft foundation system.

5.5.3 Pile Foundations

Where pile foundations are adopted through fill and reactive materials, it will be necessary to carefully assess the potential for uplift due to reactive soils/fill, where moisture content/suction change is possible, and placement of 'void former' beneath ground beams etc. may be required to minimise uplift pressure.

In pile design, it is suggested that shaft adhesion for compressive load design in the upper 2m of pile shaft (or the depth to very low strength or stronger rock, whichever is the lesser) be neglected if shrinkage away from pile shafts due to soil moisture variations is considered possible. In assessment of pile structural capacity, an ultimate (unfactored) uplift stress of not less than 100kPa in clays (and reactive fill) should be allowed for over the upper 2m of pile shaft, due to soil (or reactive fill) swell, if moisture variations are possible.

The effect of group action should be considered where piles are to act in groups, and an overall foundation capacity of less than the sum of individual pile capacities could result where piles are closely spaced.

Bored pile design could be based on the maximum allowable working end bearing and shaft adhesion values nominated in Table 8. Ultimate (failure) stress bearing pressures can be obtained by multiplying the working stress values by 2.5. Pile design stresses (and hence, pile dimensions) should be confirmed by engineering inspection at the time of construction.

Table 8: Working Shaft/End Bearing Capacity for Bored Pile Design

Material	Strength	Shaft Adhesion (kPa)	End Bearing ⁽¹⁾ (kPa)
Level 1 Controlled Fill	-	5	150
Sandy/Silty Clay	very stiff	15	200
	hard	25	300
Siltstone/Sandstone (rock)	extremely low	45	500

⁽¹⁾ For pile length to pile diameter ratio >4.0 in founding strata and no 'softer' underlying materials

5.5.4 Geotechnical Strength Reduction Factor

Application of an appropriate geotechnical strength reduction factor (ϕ_g) to $R_{d,ug}$ gives a design geotechnical strength ($R_{d,g}$) which, using limit state design, must be greater than or equal to the design action effect (E_d). A project specific geotechnical strength reduction factor value should be assessed based on the average risk rating values determined using Table 4.3.2(A) of AS2159 – 2009 *Piling – Design and Installation* in combination with an assessment of the level of redundancy in the overall piling system and the amount/type of pile testing proposed.

Values of ϕ_g could range from 0.4 for very high-risk piles with low redundancy and limited testing through to 0.9 for very low risk piles with high redundancy where not less than 10% of the piles are subject to a successful static load test. As an example, a typical ϕ_g value for moderate risk precast piles in a low redundancy pile system where 10% of all piles are subject to dynamic load testing would be approximately 0.7. Location and project specific assessment of ϕ_g using the methodology described in Section 4.3 of AS2159 – 2009 must be carried out as part of detailed pile design.

5.6 **Foundation Settlement**

It will be necessary to undertake detailed foundation settlement analysis as part of the foundation design process (adopting appropriate location specific ground conditions and properties), once footing sizes, loads and spacing are confirmed; settlement assessment should include the effects of 'group' settlement behaviour for closely spaced footings.

Additional movement could result from reactive soil/fill movement, which must be allowed for in design.

5.6.1 Strip and Pad Foundations

For properly constructed pad and strip footings sized based on the working bearing pressures given in Table 6, settlement would not be expected to exceed 1.0% to 1.5% of footing width for pad footings and 2.0% to 2.5% of footing width for strip footings, provided no 'softer' soils are present below founding level.

If the firm compressible clays are present close to proposed founding locations, then specific site settlement assessment would be required on a case-by-case basis.

5.6.2 Raft

Details of the raft size, founding level(s) and loading are required before an analysis of raft settlement can be made.

5.6.3 Bored Piles

Pile settlement (including 'group' effects) should be checked in detail as part of foundation design, however, single bored pile settlements for properly constructed piles are typically found to be in the range of 0.5% to 1.0% of pile diameter at working load, provided that 'softer' seams are not located within the zone of base stress influence.

5.7 **Pavement Subgrade Properties**

Subgrade properties will vary significantly over the site following bulk earthworks and testing will be required at the time of construction in order to confirm design values. For the purposes of initial costing and preliminary design the subgrade values given in Table 9 may be adopted. The values presented in Table 9 are based on the assumption that the subgrade is prepared in accordance with Section 5.2. Detailed subgrade testing at the time of earthworks will be required to confirm design subgrade properties.

Table 9: Preliminary Subgrade Design Values

Subgrade Material	CBR (%)	Modulus of Subgrade Reaction (kPa/mm) ⁽¹⁾
Silty Clays (natural or controlled fill)	1 – 3	10 – 30

⁽¹⁾ For transient loading only

Due to the reactive nature of the clays encountered at the site, it is suggested that on-ground slabs are fully doweled and joints between slabs sealed to minimise potential damage from surface movements caused by subgrade moisture content variations. Allowance should be made in design for the effect of differential slab movements (due to reactive soil behaviour) if slabs are cast over different subgrade types.

Reactive subgrade materials should not be allowed to 'dry out', otherwise significant softening and soil-swell movements on 'wetting up' could potentially occur.

5.8 **Earthquake Site Factor**

With reference to Australian Standard AS1170.4-2007 (R2018/Amdt2-2018) Structural design actions- Earthquake actions in Australia, it is considered that the following may be adopted for the site:

Hazard Design Factor (Z): 0.09

Class Definitions: The Class Definition will vary across the site, based on the depth of earthworks to be undertaken. The site in its current condition is considered to be a Class Ce-Shallow soil site. The appropriate Class Definition must be assessed on a location by location basis, once earthworks design has been completed.

BUTLER PARTNERS PTY LTD

NICK BLOXSOM
Associate

Reviewed by:
JENNY SALAS
Senior Geotechnical Engineer

Important Information about Your Geotechnical Engineering Report

Subsurface problems are a principal cause of construction delays, cost overruns, claims, and disputes.

While you cannot eliminate all such risks, you can manage them. The following information is provided to help.

Geotechnical Services Are Performed for Specific Purposes, Persons, and Projects

Geotechnical engineers structure their services to meet the specific needs of their clients. A geotechnical engineering study conducted for a civil engineer may not fulfill the needs of a construction contractor or even another civil engineer. Because each geotechnical engineering study is unique, each geotechnical engineering report is unique, prepared *solely* for the client. No one except you should rely on your geotechnical engineering report without first conferring with the geotechnical engineer who prepared it. *And no one — not even you — should apply the report for any purpose or project except the one originally contemplated.*

Read the Full Report

Serious problems have occurred because those relying on a geotechnical engineering report did not read it all. Do not rely on an executive summary. Do not read selected elements only.

A Geotechnical Engineering Report Is Based on A Unique Set of Project-Specific Factors

Geotechnical engineers consider a number of unique, project-specific factors when establishing the scope of a study. Typical factors include: the client's goals, objectives, and risk management preferences; the general nature of the structure involved, its size, and configuration; the location of the structure on the site; and other planned or existing site improvements, such as access roads, parking lots, and underground utilities. Unless the geotechnical engineer who conducted the study specifically indicates otherwise, do not rely on a geotechnical engineering report that was:

- not prepared for you,
- not prepared for your project,
- not prepared for the specific site explored, or
- completed before important project changes were made.

Typical changes that can erode the reliability of an existing geotechnical engineering report include those that affect:

- the function of the proposed structure, as when it's changed from a parking garage to an office building, or from a light industrial plant to a refrigerated warehouse,

- elevation, configuration, location, orientation, or weight of the proposed structure,
- composition of the design team, or
- project ownership.

As a general rule, *always* inform your geotechnical engineer of project changes—even minor ones—and request an assessment of their impact. *Geotechnical engineers cannot accept responsibility or liability for problems that occur because their reports do not consider developments of which they were not informed.*

Subsurface Conditions Can Change

A geotechnical engineering report is based on conditions that existed at the time the study was performed. *Do not rely on a geotechnical engineering report* whose adequacy may have been affected by: the passage of time; by man-made events, such as construction on or adjacent to the site; or by natural events, such as floods, earthquakes, or groundwater fluctuations. *Always* contact the geotechnical engineer before applying the report to determine if it is still reliable. A minor amount of additional testing or analysis could prevent major problems.

Most Geotechnical Findings Are Professional Opinions

Site exploration identifies subsurface conditions only at those points where subsurface tests are conducted or samples are taken. Geotechnical engineers review field and laboratory data and then apply their professional judgment to render an opinion about subsurface conditions throughout the site. Actual subsurface conditions may differ—sometimes significantly—from those indicated in your report. Retaining the geotechnical engineer who developed your report to provide construction observation is the most effective method of managing the risks associated with unanticipated conditions.

A Report's Recommendations Are *Not* Final

Do not overrely on the construction recommendations included in your report. *Those recommendations are not final*, because geotechnical engineers develop them principally from judgment and opinion. Geotechnical engineers can finalize their recommendations only by observing actual

subsurface conditions revealed during construction. *The geotechnical engineer who developed your report cannot assume responsibility or liability for the report's recommendations if that engineer does not perform construction observation.*

A Geotechnical Engineering Report Is Subject to Misinterpretation

Other design team members' misinterpretation of geotechnical engineering reports has resulted in costly problems. Lower that risk by having your geotechnical engineer confer with appropriate members of the design team after submitting the report. Also retain your geotechnical engineer to review pertinent elements of the design team's plans and specifications. Contractors can also misinterpret a geotechnical engineering report. Reduce that risk by having your geotechnical engineer participate in prebid and preconstruction conferences, and by providing construction observation.

Do Not Redraw the Engineer's Logs

Geotechnical engineers prepare final boring and testing logs based upon their interpretation of field logs and laboratory data. To prevent errors or omissions, the logs included in a geotechnical engineering report should *never* be redrawn for inclusion in architectural or other design drawings. Only photographic or electronic reproduction is acceptable, *but recognize that separating logs from the report can elevate risk.*

Give Contractors a Complete Report and Guidance

Some owners and design professionals mistakenly believe they can make contractors liable for unanticipated subsurface conditions by limiting what they provide for bid preparation. To help prevent costly problems, give contractors the complete geotechnical engineering report, *but* preface it with a clearly written letter of transmittal. In that letter, advise contractors that the report was not prepared for purposes of bid development and that the report's accuracy is limited; encourage them to confer with the geotechnical engineer who prepared the report (a modest fee may be required) and/or to conduct additional study to obtain the specific types of information they need or prefer. A prebid conference can also be valuable. *Be sure contractors have sufficient time* to perform additional study. Only then might you be in a position to give contractors the best information available to you, while requiring them to at least share some of the financial responsibilities stemming from unanticipated conditions.

Read Responsibility Provisions Closely

Some clients, design professionals, and contractors do not recognize that geotechnical engineering is far less exact than other engineering disciplines. This lack of understanding has created unrealistic expectations that

have led to disappointments, claims, and disputes. To help reduce the risk of such outcomes, geotechnical engineers commonly include a variety of explanatory provisions in their reports. Sometimes labeled "limitations" many of these provisions indicate where geotechnical engineers' responsibilities begin and end, to help others recognize their own responsibilities and risks. *Read these provisions closely.* Ask questions. Your geotechnical engineer should respond fully and frankly.

Geoenvironmental Concerns Are Not Covered

The equipment, techniques, and personnel used to perform a *geoenvironmental* study differ significantly from those used to perform a *geotechnical* study. For that reason, a geotechnical engineering report does not usually relate any geoenvironmental findings, conclusions, or recommendations; e.g., about the likelihood of encountering underground storage tanks or regulated contaminants. *Unanticipated environmental problems have led to numerous project failures.* If you have not yet obtained your own geoenvironmental information, ask your geotechnical consultant for risk management guidance. *Do not rely on an environmental report prepared for someone else.*

Obtain Professional Assistance To Deal with Mold

Diverse strategies can be applied during building design, construction, operation, and maintenance to prevent significant amounts of mold from growing on indoor surfaces. To be effective, all such strategies should be devised for the *express purpose* of mold prevention, integrated into a comprehensive plan, and executed with diligent oversight by a professional mold prevention consultant. Because just a small amount of water or moisture can lead to the development of severe mold infestations, a number of mold prevention strategies focus on keeping building surfaces dry. While groundwater, water infiltration, and similar issues may have been addressed as part of the geotechnical engineering study whose findings are conveyed in this report, the geotechnical engineer in charge of this project is not a mold prevention consultant; ***none of the services performed in connection with the geotechnical engineer's study were designed or conducted for the purpose of mold prevention. Proper implementation of the recommendations conveyed in this report will not of itself be sufficient to prevent mold from growing in or on the structure involved.***

Rely, on Your ASFE-Member Geotechnical Engineer for Additional Assistance

Membership in ASFE/THE BEST PEOPLE ON EARTH exposes geotechnical engineers to a wide array of risk management techniques that can be of genuine benefit for everyone involved with a construction project. Confer with your ASFE-member geotechnical engineer for more information.



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PROJECT
PROJECTS OFFICE

PROJECT ADDRESS
Stanwell Power Station, Stanwell

CLIENT

Stanwell Corporation Limited



REV.	DATE	ISSUE	APP.
A	20/06/2025	FINAL	DH

DRAWING
Bore Locations

REPORT
-

PROJECT No.
RX25-1156A

DRAWING No.
1.02



LEGEND

- B1 Bores (Geotechnical and Environmental)
- B2 Bores (Environmental)
- Site Boundary

APPENDIX A

BORE REPORT SHEETS WITH EXPLANATORY NOTES

BORE REPORT

Client: Stanwell Corporation Ltd
Project: Proposed Projects Office
Location: Stanwell Power Station, Stanwell
Project No: RX25-1156A

BORE 1

Page No: 1 of 1
Date: 29 May 2025
Ground Surface Level:

Depth (m)	Description	RL (m)	Lithology	Sample Type	Sample Depth (m)	Test Results
0	FILL - silty gravel, dark grey, fine to medium size sub-angular gravel	0.0			0.1	
	SILTY CLAY (CL) - very stiff, brown, with fine to coarse grained sand, trace fine to coarse size sub-angular gravel			D	0.5	
1		-1.0		S	0.95	10,10,16 N=26
				D	1.5	
2		-2.0		S	1.95	11,21,28 N=49
				D		
3	SANDY CLAY (CL) - hard, brown, with fine to coarse grained sand, trace fine to medium size sub-angular gravel	-3.0		S	3.0	23,30/120mm
4		-4.0		D	3.27	
	SILTSTONE/SANDSTONE (XW) - extremely low strength, grey brown			S	4.5	30,30/40mm
5		-5.0		D	4.69	
6		-6.0		S	6.0	30/120mm
	End of Bore at 6.12 m	-6.12			6.12	
7		-7.0				
8		-8.0				
9		-9.0				
10		-10.0				

D Disturbed Sample	S Standard Penetration Test (SPT)	E Environmental Sample	Is(50) Point Load Test Result (MPa)
U Undisturbed Tube (50mm dia)	HB SPT Hammer Bouncing	Up Pushtube Sample	(d) Diametral Test
B Bulk Sample	() No Sample Recovery	C NMLC Coring	(a) Axial Test
pp Pocket Penetrometer Test (kPa)	V Vane Shear Strength, Uncorrected (kPa)		(i) Lump Test

Rig: Jacro 350

Drilling Method: Auger

Groundwater: No free groundwater encountered during auger drilling

Remarks:

Logged by: SC/SR

Co-Ordinates E: 226467

Co-Ordinates N: 7397692

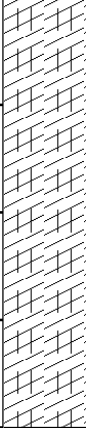
ENVIRONMENTAL BORE REPORT



Client: Stanwell Corporation Ltd
Project: Proposed Projects Office
Location: Stanwell Power Station, Stanwell
Project No: RX25-1156A

BORE 2

Page No: 1 of 1
Date: 29 May 2025
Ground Surface Level: RL 62.0m*

Depth (m)	Description	RL (m)	Lithology	Sample Type	Sample Depth (m)	Sample ID	Test Results
0	FILL - silty gravel, dark grey, fine to medium grained sized sub-angular gravel - silty clay, brown SILTY CLAY - pale brown, with fine to medium grained sized sub-angular gravel	62.0		EA	0.0	B2-0	
				EA	0.025	B2-1	
					0.1		
				EA	0.2		
					0.3		
				EA	0.4	B2-2	
				EA	0.8	B2-4	
1	End of Bore at 1 m	61.0			1.0		
2		60.0					

D Disturbed Sample
A Asbestos Sample

E Environmental Sample
() No Sample Recovery

C NMLC Coring
P PFAS Sample

This Environmental Bore Report is not intended for geotechnical engineering interpretation

Rig: Jacro 350

Drilling Method: Auger

Groundwater: No free groundwater encountered during auger drilling

Remarks: *Approximate ground surface levels interpolated from Rockhampton Regional Council's interactive mapping database

Logged by: DH

Co-Ordinates E: 226462

Co-Ordinates N: 7397706

ENVIRONMENTAL BORE REPORT



Client: Stanwell Corporation Ltd
Project: Proposed Projects Office
Location: Stanwell Power Station, Stanwell
Project No: RX25-1156A

BORE 3

Page No: 1 of 1
Date: 29 May 2025
Ground Surface Level: RL 62.0m*

Depth (m)	Description	RL (m)	Lithology	Sample Type	Sample Depth (m)	Sample ID	Test Results
0	FILL - silty gravel, dark grey, fine to medium sized sub-angular gravel - silty clay, brown SILTY CLAY - pale brown, with fine to coarse sized sub-angular gravel	62.0		E/A	0.0	B3-0	
				E/A	0.025	B3-1	
					0.1		
					0.2		
				E/A	0.3	B3-2	
					0.4	QC01	
					0.8		
				E/A	1.0	B3-3	
1		61.0			1.2		
	- pale brown, with fine to coarse grained sand			E/A	1.3	B3-4	
					1.5		
				E/A	1.8	B3-5	
2	End of Bore at 2 m		60.0				

D Disturbed Sample
A Asbestos Sample

E Environmental Sample
() No Sample Recovery

C NMLC Coring
P PFAS Sample

This Environmental Bore Report is not intended for geotechnical engineering interpretation

Rig: Jacro 350

Drilling Method: Auger

Groundwater: No free groundwater encountered during auger drilling

Remarks: *Approximate ground surface levels interpolated from Rockhampton Regional Council's interactive mapping database

Logged by: DH

Co-Ordinates E: 226440

Co-Ordinates N: 7397706

BORE REPORT



Client: Stanwell Corporation Ltd
Project: Proposed Projects Office
Location: Stanwell Power Station, Stanwell
Project No: RX25-1156A

BORE 4

Page No: 1 of 1
Date: 30 May 2025
Ground Surface Level:

Depth (m)	Description	RL (m)	Lithology	Sample Type	Sample Depth (m)	Test Results
0	FILL - silty sand, brown, fine to coarse grained sand, with fine to medium size gravel	0.0				
0.2				D	0.2	
0.5				S	0.5	
1	SILTY CLAY (CL) - hard, pale brown, with fine to coarse grained sand, trace fine to coarse size sub-angular gravel	-1.0		D	0.95	13,13,20 N=33
1.5				S	1.5	
1.95		-2.0		D	1.95	11,21,28 N=49
2						
3	- brown mottled grey	-3.0		S	3.0	23,30/120mm
3.27					3.27	
4		-4.0		D		
4.5	SILTSTONE/SANDSTONE (XW) - extremely low strength, brown mottled grey			S	4.5	17,30/105mm
4.755		-5.0			4.755	
5				D	5.0	
6		-6.0		S	6.0	30,30/100mm
6.25	End of Bore at 6.25 m	-6.25			6.25	
7		-7.0				
8		-8.0				
9		-9.0				
10		-10.0				

D Disturbed Sample	S Standard Penetration Test (SPT)	E Environmental Sample	Is(50) Point Load Test Result (MPa)
U Undisturbed Tube (50mm dia)	HB SPT Hammer Bouncing	Up Pushtube Sample	(d) Diametral Test
B Bulk Sample	() No Sample Recovery	C NMLC Coring	(a) Axial Test
pp Pocket Penetrometer Test (kPa)	V Vane Shear Strength, Uncorrected (kPa)		(i) Lump Test

Rig: Jacro 350

Drilling Method: Auger

Groundwater: No free groundwater encountered during auger drilling

Remarks:

Logged by: SC/SR

Co-Ordinates E: 226461

Co-Ordinates N: 7397735

Notes on Description and Classification of Soil

The methods of description and classification of soils used in this report are generally based on Australian Standard AS1726-1993 Geotechnical Site Investigations.

Soil description is based on an assessment of disturbed samples, as recovered from bores and excavations, or from undisturbed materials as seen in excavations and exposures or in undisturbed samples. Descriptions given on report sheets are an interpretation of the conditions encountered at the time of investigation.

In the case of cone or piezocone penetrometer tests, actual soil samples are not recovered and soil description is inferred based on published correlations, past experience and comparison with bore and/or test pit data (if available).

Soil classification is based on the particle size distribution of the soil and the plasticity of the portion of the material finer than 0.425mm. The description of particle size distribution and plasticity is based on the results of visual field estimation, laboratory testing or both. When assessed in the field, the properties of the soil are estimated; precise description will always require laboratory testing to define soil properties.

Where soil can be clearly identified as FILL this will be noted as the main soil type followed by a description of the composition of the fill (e.g. FILL – yellow-brown, fine to coarse grained gravelly clay fill with concrete rubble). If the soil is assessed as possibly being fill this will be noted as an additional observation.

Soils are generally described using the following sequence of terms. In certain instances, not all of the terms will be included in the soil description.

MAIN SOIL TYPE (CLASSIFICATION GROUP SYMBOL)

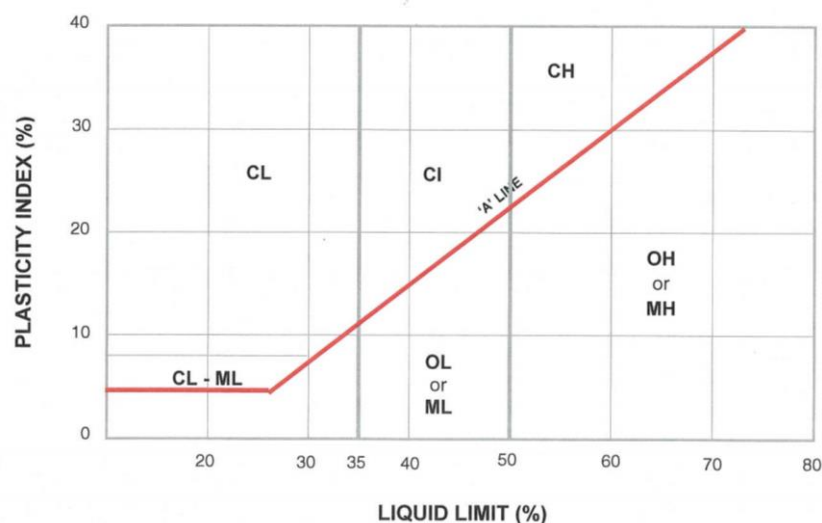
- strength/density, colour, structure/grain size, secondary and minor components, additional observations

Information on the definition of descriptive and classification terms follows.

SOIL TYPE and CLASSIFICATION GROUP SYMBOLS

	Major Divisions	Particle Size	Classification Group Symbol	Typical Names
COARSE GRAINED SOILS (more than half of material is larger than 0.075mm)	BOULDERS	>200mm		
	COBBLES	63 – 200mm		
	GRAVELS (more than half of coarse fraction is larger than 2.36mm)	Coarse: 20 – 63mm Medium: 6 – 20mm Fine: 2.36 – 6mm	GW	Well graded gravels, gravel-sand mixtures, little or no fines.
			GP	Poorly graded gravels and gravel-sand mixtures, little or no fines, uniform gravels.
			GM	Silty gravels, gravel-sand-silt mixtures.
			GC	Clayey gravels, gravel-sand-clay mixtures.
	SANDS (more than half of coarse fraction is smaller than 2.36mm)	Coarse: 0.6 – 2.36mm Medium: 0.2 – 0.6mm Fine: 0.075 – 0.2mm	SW	Well graded sands, gravelly sands, little or no fines.
			SP	Poorly graded sands and gravelly sands; little or no fines, uniform sands.
			SM	Silty sands, sand-silt mixtures.
			SC	Clayey sands, sand-clay mixtures.
FINE GRAINED SOILS (more than half of material is smaller than 0.075mm)	SILTS & CLAYS (liquid limit <50%)		ML	Inorganic silts and very fine sands, silty/clayey fine sands or clayey silts with low plasticity.
			CL and CI	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays.
			OL	Organic silts and organic silty clays of low plasticity.
	SILTS & CLAYS (liquid limit >50%)		MH	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils.
			CH	Inorganic clays of high plasticity.
			OH	Organic clays of medium to high plasticity, organic silts.
	HIGHLY ORGANIC SOILS		Pt	Peat and other highly organic soils.

PLASTICITY CHART FOR CLASSIFICATION OF FINE GRAINED SOILS



(Reference: Australian Standard AS1726-1993 Geotechnical site investigations)

DESCRIPTIVE TERMS FOR MATERIAL PROPORTIONS

Coarse Grained Soils		Fine Grained Soils	
% Fines	Modifier	% Coarse	Modifier
<5	Omit, or use 'trace'	<15	Omit, or use trace.
5 – 12	Describe as 'with clay/silt' as applicable.	15 – 30	Describe as 'with sand/gravel' as applicable.
>12	Prefix soil as 'silty/clayey' as applicable	>30	Prefix soil as 'sandy/gravelly' as applicable.

STRENGTH TERMS – COHESIVE SOILS

Strength Term	Undrained Shear Strength	Field Guide to Strength
Very soft	<12kPa	Exudes between the fingers when squeezed in hand.
Soft	12 – 25kPa	Can be moulded by light finger pressure.
Firm	25 – 50kPa	Can be moulded by strong finger pressure.
Stiff	50 – 100kPa	Cannot be moulded by fingers, can be indented by thumb.
Very stiff	100 – 200kPa	Can be indented by thumb nail.
Hard	>200kPa	Can be indented with difficulty by thumb nail.

DENSITY TERMS – NON COHESIVE SOILS

Density Term	Density Index	SPT "N"	CPT Cone Resistance
Very loose	<15%	0 – 5	0 – 2MPa
Loose	15 – 35%	5 – 10	2 – 5MPa
Medium dense	35 – 65%	10 – 30	5 – 15MPa
Dense	65 – 85%	30 – 50	15 – 25MPa
Very dense	>85%	>50	>25MPa

COLOUR

The colour of a soil will generally be described in a 'moist' condition using simple colour terms (e.g. black, grey, red, brown etc.) modified as necessary by "pale", "dark", "light" or "mottled". Borderline colours will be described as a combination of colours (e.g. grey-brown).

EXAMPLE

e.g. CLAYEY SAND (SC) – medium dense, grey-brown, fine to medium grained with silt.

Indicates a medium dense, grey-brown, fine to medium grained clayey sand with silt.

Notes on Description and Classification of Rock

The methods of description and classification of rock used in this report are generally based on Australian Standard AS1726-1993 *Geotechnical site investigations*.

Rock description is based on an assessment of disturbed samples, as recovered from bores and excavations, or from undisturbed materials as seen in excavations and exposures, or in core samples. Descriptions given on report sheets are an interpretation of the conditions encountered at the time of investigation.

Notes outlining the method and terminology adopted for the description of rock defects are given below, however, detailed information on defects can generally only be determined where rock core is taken, or excavations or exposures allow detailed observation and measurement.

Rocks are generally described using the following sequence of terms. In certain instances not all of the terms will be included in the rock description.

ROCK TYPE (WEATHERING SYMBOL), strength, colour, grain size, defect frequency

Information on the definition of descriptive and classification terms follows.

ROCK TYPE

In general, simple rock names are used rather than precise geological classifications.

ROCK MATERIALS WEATHERING CLASSIFICATION

Term	Weathering Symbol	Definition
Residual soil	RS	Soil developed from extremely weathered rock; the mass structure and substance fabrics are no longer evident; there is a large change in volume but the soil has not been significantly transported.
Extremely weathered	XW	Rock is weathered to such an extent that it has 'soil' properties, i.e. it either disintegrates or can be remoulded in water.
Distinctly weathered *	DW	Rock strength usually changed by weathering. The rock may be highly discoloured, usually by ironstaining. Porosity may be increased by leaching, or may be decreased due to deposition of weathering products in pores.
- Highly weathered	HW	Rock substance affected by weathering to the extent that limonite staining or bleaching affects the whole of the rock substance and other signs of chemical or physical decomposition are evident. Porosity and strength may be increased or decreased compared to the fresh rock, usually as a result of iron leaching or deposition. The colour and strength of the original fresh rock substance is no longer recognisable.
- Moderately weathered	MW	Rock substance affected by weathering to the extent that staining extends throughout the whole of the rock substance and the original colour of the fresh rock may be no longer recognisable.
Slightly weathered	SW	Rock is slightly discoloured but shows little or no change of strength from fresh rock.
Fresh	FR	Rock shows no sign of decomposition or staining.

* Subdivision of this weathering grade into highly and moderately may be used where applicable.

STRENGTH OF ROCK MATERIAL

Term	Symbol	Point Load Index I_s (50)	Field Guide To Strength
Extremely low	EL	<0.03MPa	Easily remoulded by hand to a material with soil properties.
Very low	VL	0.03 – 0.1MPa	Material crumbles under firm blows with sharp end of pick; can be peeled with knife; too hard to cut a triaxial sample by hand. Pieces up to 30mm thick can be broken by finger pressure.
Low	L	0.1 – 0.3MPa	Easily scored with a knife; indentations 1mm to 3mm show in the specimen with firm blows of the pick point; has dull sound under hammer. A piece of core 150mm long 50mm diameter may be broken by hand. Sharp edges of core may be friable and break during handling.
Medium	M	0.3 – 1.0MPa	Readily scored with a knife; a piece of core 150mm long by 50mm diameter can be broken by hand with difficulty.
High	H	1.0 – 3.0MPa	A piece of core 150mm long by 50mm diameter cannot be broken by hand but can be broken by a pick with a single firm blow; rock rings under hammer.
Very high	VH	3.0 – 10.0MPa	Hand specimen breaks with pick after more than one blow; rock rings under hammer.
Extremely high	EH	>10MPa	Specimen requires many blows with geological pick to break through intact material; rock rings under hammer.

Notes:

- These terms refer to the strength of the rock material and not to the strength of the rock mass which may be considerably weaker due to the effect of rock defects.
- The field guide visual assessment for rock strength may be used for preliminary assessment or when point load testing is not available.
- Anisotropy of rock may affect the field assessment of strength.

COLOUR

The colour of a rock will generally be described in a 'moist' condition using simple colour terms (e.g. black, grey, red, brown, etc) modified as necessary by 'pale', 'dark', 'light' or 'mottled'. Borderline colours will be described as a combination of colours (e.g. grey-brown).

GRAIN SIZE

Descriptive Term	Particle Size Range
Coarse grained	0.6 – 2.0mm
Medium grained	0.2 – 0.6mm
Fine grained	0.06 – 0.2mm

DEFECT FREQUENCY

Where appropriate, a defect frequency may be recorded as part of the rock description and will be expressed as the number of natural (or interpreted natural) defects present in an equivalent one metre length of core; by use of the following defect frequency descriptive terms; or both. The descriptive terms refer to the spacing of all types of natural defects along which the rock is discontinuous and include, bedding plane partings, joints and other rock defects, but excludes known artificial fractures such as drilling breaks.

Defect Frequency	Description
Fragmented	Rock core is comprised primarily of fragments of length less than 20mm, and mostly of width less than the core diameter.
Highly Fractured	Core lengths are generally less than 20mm to 40mm with occasional fragments.
Fractured	Core lengths are mainly 30mm to 100mm with occasional shorter and longer sections.
Fractured to Slightly Fractured	Core lengths are mainly 100mm to 300mm with occasional shorter to longer sections.
Slightly Fractured	Core lengths are generally 300mm to 1,000mm with occasional longer sections and occasional sections of 100mm to 300mm.
Unbroken	The core does not contain any fractures.

EXAMPLE

e.g. SANDSTONE (XW) – low strength, pale brown, fine to coarse grained, slightly fractured.

ROCK DEFECT LOGGING

Defects are discontinuities in the rock mass and include joints, sheared zones, cleavages and bedding partings. The ability to observe and log defects will depend on the investigation methodology. Defects logged in core are described using the abbreviations noted in the following tables.

The *depth* noted in the description is measured in metres from the ground surface, the *defect angle* is measured in degrees from horizontal, and the defect *thickness* is measured normal to the plane of the defect and is in millimetres (unless otherwise noted).

Defects are generally described using the following sequence of terms:

Depth, Defect Type, Defect Angle (dip), Surface Roughness, Infill, Thickness

DEFECT TYPE

B	– Bedding
J	– Joint
S	– Shear Zone
C	– Crushed Zone

SURFACE ROUGHNESS

i	- rough or irregular, stepped
ii	- smooth, stepped
iii	- slickensided, stepped
iv	- rough or irregular, undulating
v	- smooth, undulating
vi	- slickensided, undulating
vii	- rough or irregular, planar
viii	- smooth planar
ix	- slickensided, planar

INFILL

Infill refers to secondary minerals or other materials formed on the surface of the defect and some common descriptions are given in the following table together with their abbreviations.

Ls	- limonite staining
Fe	- iron staining
Cl	- clay
Mn	- manganese staining
Qtz	- quartz
Ca	- calcite
Clean	- no visible infill

EXAMPLE

3.59m, J, 90, vii, Ls, 1mm

indicates a joint at 3.59m depth that is at 90° to horizontal (i.e. vertical), is rough or irregular and planar, limonite stained and 1mm thick.

APPENDIX B

LABORATORY TEST RESULT REPORT SHEETS

Material Test Report

Report Number: RX25-1156A-1B
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157A
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 01/07/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 1, Depth: 0.1-0.5



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246 Kent Street Rockhampton QLD 4700
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Email: abird@butlerpartners.com.au



Accredited for compliance with ISO/IEC 17025 - Testing

Approved Signatory: Adam Bird
Laboratory Manager
NATA Accredited Laboratory Number: 19529

Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	4 *		
Soil Description	Silty Clay		
Nature of Water	Distilled		
* Mineral Present	Carbonate		

Material Test Report

Report Number: RX25-1156A-1D
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157A
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 02/07/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 1, Depth: 0.1-0.5



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NATA Accredited Laboratory Number: 19529

pH Value of Soil (AS 1289 4.3.1)		Min	Max
Soil-suspension Temperature (°C)	20		
Distilled Water pH	7.0		
Depth	0.1-0.5		
Material Description	Silty Clay		
Moisture Condition	Air-dried		
pH	8.6		
Electrical Conductivity (µS/cm)	170.7		
For Conductivity - 1 dS/m = 1 mS/cm = 1000 µS/cm			
Electrical Conductivity not covered by accreditation.			

Material Test Report

Report Number: RX25-1156A-1B
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157E
Date Sampled: 30/05/2025
Dates Tested: 16/06/2025 - 01/07/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 4, Depth: 0.2-0.5



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Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	4 *		
Soil Description	Silty Clay		
Nature of Water	Distilled		
* Mineral Present	Carbonate		

Material Test Report

Report Number: RX25-1156A-1D
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157E
Date Sampled: 30/05/2025
Dates Tested: 16/06/2025 - 02/07/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 4, Depth: 0.2-0.5



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pH Value of Soil (AS 1289 4.3.1)		Min	Max
Soil-suspension Temperature (°C)	20		
Distilled Water pH	7.0		
Depth	0.2-0.5		
Material Description	Silty Clay		
Moisture Condition	Air-dried		
pH	9.2		
Electrical Conductivity (µS/cm)	455.3		
For Conductivity - 1 dS/m = 1 mS/cm = 1000 µS/cm			
Electrical Conductivity not covered by accreditation.			

Material Test Report

Report Number: RX25-1156A-1A
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157B
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 25/06/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 1, Depth: 0.95-1.5



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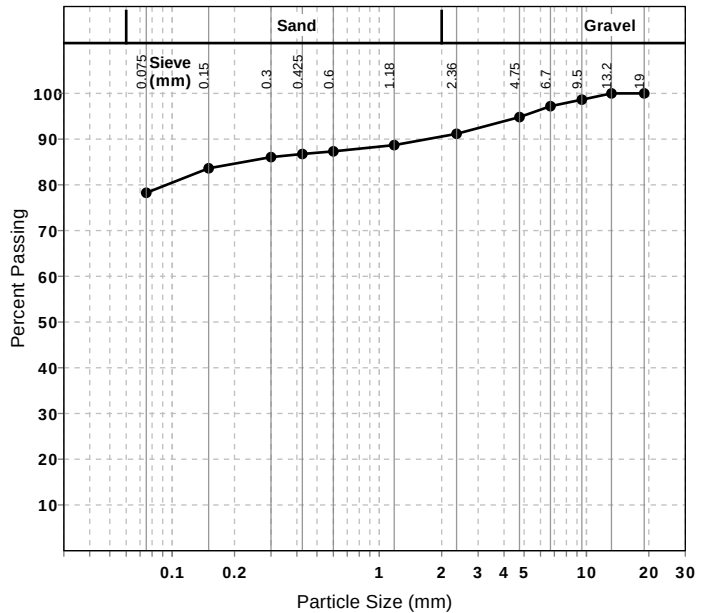
Particle Size Distribution (AS1289 3.6.1 & 2.1.1 & Q253)

Sieve	Passed %	Passing Limits
19 mm	100	
13.2 mm	100	
9.5 mm	99	
6.7 mm	97	
4.75 mm	95	
2.36 mm	91	
1.18 mm	89	
0.6 mm	87	
0.425 mm	87	
0.3 mm	86	
0.15 mm	84	
0.075 mm	78	

Particle size and shape properties of a soil (Q253)		Min	Max
Coefficient of Uniformity (D_{60}/D_{10})	708.0		
Fines Ratio	0.90		
Fines to Sand Ratio	0.86		
% < 0.075 mm / % < 0.300 mm Ratio	0.91		
Coefficient of Curvature	0.32		
Grading Coefficient			

Moisture Content (AS 1289 2.1.1)		Min	Max
Moisture Content (%)	12.9		

Particle Size Distribution



Material Test Report

Report Number: RX25-1156A-1A
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157C
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 30/06/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 1, Depth: 3.27-4.5



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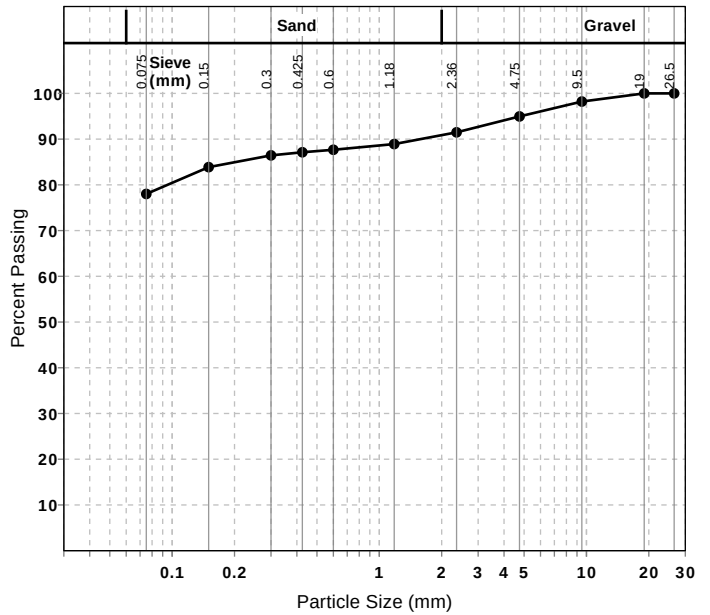
Approved Signatory: Adam Bird
Laboratory Manager
NATA Accredited Laboratory Number: 19529

Particle Size Distribution (AS1289 3.6.1)

Sieve	Passed %	Passing Limits	
26.5 mm	100		
19 mm	100		
9.5 mm	98		
4.75 mm	95		
2.36 mm	91		
1.18 mm	89		
0.6 mm	88		
0.425 mm	87		
0.3 mm	86		
0.15 mm	84		
0.075 mm	78		

		Min	Max
Coefficient of Uniformity (D_{60}/D_{10})		446.5	
Moisture Content (AS 1289 2.1.1)		Min	Max
Moisture Content (%)		14.4	

Particle Size Distribution



Material Test Report

Report Number: RX25-1156A-1A
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157F
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 20/06/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 4, Depth: 0.95-1.5



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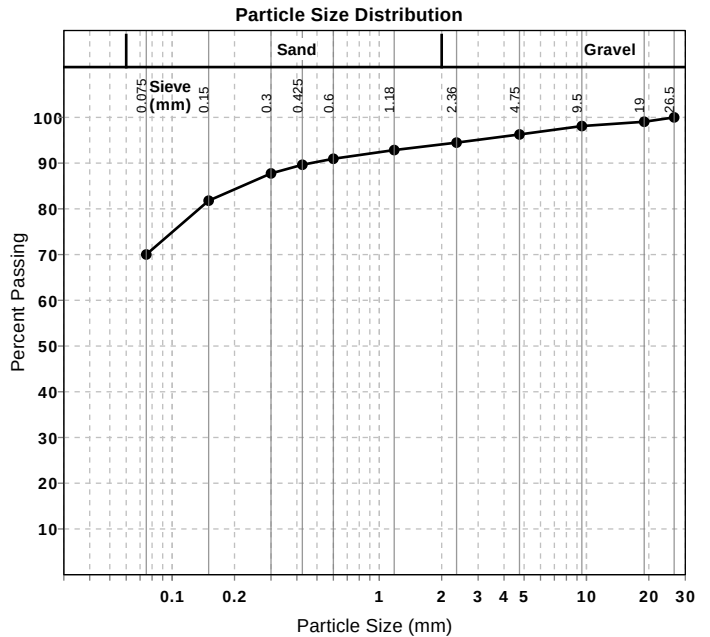
NATA Accredited Laboratory Number: 19529

Particle Size Distribution (AS1289 3.6.1 & 2.1.1 & Q253)

Sieve	Passed %	Passing Limits
26.5 mm	100	
19 mm	99	
9.5 mm	98	
4.75 mm	96	
2.36 mm	94	
1.18 mm	93	
0.6 mm	91	
0.425 mm	90	
0.3 mm	88	
0.15 mm	82	
0.075 mm	70	

Particle size and shape properties of a soil (Q253)		Min	Max
Coefficient of Uniformity (D_{60}/D_{10})	19.0		
Fines Ratio	0.78		
Fines to Sand Ratio	0.89		
% < 0.075 mm / % < 0.300 mm Ratio	0.92		
Coefficient of Curvature	0.55		
Grading Coefficient	5.3		

Moisture Content (AS 1289 2.1.1)		Min	Max
Moisture Content (%)	11.3		



Material Test Report

Report Number: RX25-1156A-1C
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157B
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 30/06/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 1, Depth: 0.95-1.5



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Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1 & Q252)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Passing 0.425 (%)	87		
Liquid Limit (%)	69		
Plastic Limit (%)	18		
Plasticity Index (%)	51		
Weighted Plasticity Index (%)	4424		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Moisture Condition Determined By	AS 1289.3.1.2		
Linear Shrinkage (%)	9.5		
Weighted Linear Shrinkage (%)	831		
Cracking Crumbling Curling	None		

Material Test Report

Report Number: RX25-1156A-1C
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157C
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 30/06/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 1, Depth: 3.27-4.5



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Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1 & Q252)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Passing 0.425 (%)	87		
Liquid Limit (%)	60		
Plastic Limit (%)	16		
Plasticity Index (%)	44		
Weighted Plasticity Index (%)	3833		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Moisture Condition Determined By	AS 1289.3.1.2		
Linear Shrinkage (%)	15.0		
Weighted Linear Shrinkage (%)	1311		
Cracking Crumbling Curling	None		

Material Test Report

Report Number: RX25-1156A-1C
Issue Number: 1
Date Issued: 10/12/2025
Client: Stanwell Corporation Ltd
Level 3, 180 Ann Street, Brisbane QLD 4000
Contact: Brandon Ross
Project Number: RX25-1156A
Project Name: Proposed Projects Office
Project Location: Stanwell Power Station
Work Request: 26157
Sample Number: R25-26157F
Date Sampled: 29/05/2025
Dates Tested: 16/06/2025 - 30/06/2025
Sampling Method: AS 1289.1.2.1 6.5.3 - Power auger drilling
Preparation Method: In accordance with the test method
Sample Location: Bore 4, Depth: 0.95-1.5



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Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1 & Q252)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Passing 0.425 (%)	90		
Liquid Limit (%)	43		
Plastic Limit (%)	16		
Plasticity Index (%)	27		
Weighted Plasticity Index (%)	2420		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Moisture Condition Determined By	AS 1289.3.1.2		
Linear Shrinkage (%)	11.5		
Weighted Linear Shrinkage (%)	1033		
Cracking Crumbling Curling	None		